

AL DHAFRA, United Arab Emirates

WMO: 412162

Lat: 24.248N

Lon: 54.548E

Elev: 23

StdP: 101.05

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.9	(c) 12.0	(d) -1.7	(e) 3.3	(f) 31.6	(g) 0.8	(h) 4.0	(i) 32.0	(j) 10.1	(k) 22.2	(l) 9.2	(m) 21.6	(n) 1.9	(o) 230	(p) 0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.1	(c) 46.0	(d) 23.1	(e) 45.0	(f) 23.1	(g) 43.9	(h) 23.5	(i) 30.4	(j) 35.6	(k) 29.7	(l) 35.2	(m) 29.1	(n) 34.9	(o) 3.8	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	25.9	33.7	28.2	24.5	33.4	27.8	23.9	33.3	102.4	36.4	99.2	34.7	95.7	34.7	32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.8	(b) 8.7	(c) 7.8	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.7	19.1	21.0	23.7	28.6	32.4	34.4	36.5	37.4	34.4	30.2	25.3	20.8	(6)
(7)		DBStd	6.70	2.17	3.04	3.51	2.92	2.64	2.20	2.34	1.70	1.85	1.84	2.26	2.04	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	27	16	6	1	0	0	0	0	0	0	0	0	4	(9)
(10)		CDD10.0	6822	282	307	426	559	694	732	821	849	731	628	460	334	(10)
(11)		CDD18.3	3807	39	79	168	309	436	482	562	591	481	369	210	79	(11)
(12)		CDH23.3	55227	314	692	1697	3915	6566	7677	9374	9832	7483	5027	2108	539	(12)
(13)		CDH26.7	36489	61	264	801	2279	4372	5358	6898	7352	5117	2966	908	112	(13)
(14)	Wind	WSAvg	3.4	3.1	3.8	3.9	3.7	3.7	3.8	3.8	3.7	3.2	2.8	2.8	2.9	(14)
(15)	Precipitation	PrecAvg	58	11	11	13	6	1	1	2	1	0	0	2	9	(15)
(16)		PrecMax	145	53	89	58	54	26	16	20	22	3	6	21	66	(16)
(17)		PrecMin	3	0	1	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	43	16	18	17	13	5	3	5	4	1	1	6	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.1	35.0	38.8	41.9	45.8	46.9	47.1	47.1	45.0	40.8	36.3	31.0	(19)
(20)			MCWB	17.3	16.4	18.1	19.7	20.9	21.8	23.6	23.7	21.8	21.0	19.7	18.4	(20)
(21)		2%	DB	28.2	32.3	35.8	40.1	43.9	45.1	45.9	46.0	43.1	39.1	34.8	29.2	(21)
(22)			MCWB	16.9	17.4	18.3	19.3	21.3	22.3	23.9	23.4	22.4	21.3	19.6	18.1	(22)
(23)		5%	DB	26.8	29.9	33.8	38.3	42.2	43.8	44.2	44.9	41.9	38.0	33.0	28.0	(23)
(24)			MCWB	16.4	16.9	17.8	19.4	21.3	22.5	24.5	23.7	23.1	21.8	20.4	18.7	(24)
(25)		10%	DB	25.0	27.8	31.2	36.8	40.8	42.0	42.9	43.2	40.2	36.8	31.8	26.2	(25)
(26)			MCWB	16.2	16.5	17.7	19.3	21.6	23.1	24.8	24.3	24.0	22.1	20.3	18.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	21.1	22.1	23.6	27.7	29.2	31.2	31.2	30.6	28.1	25.3	21.8	(27)
(28)			MCDWB	24.2	25.9	28.0	30.6	36.3	35.1	37.0	36.5	34.4	33.0	29.2	26.1	(28)
(29)		2%	WB	19.0	19.9	21.1	22.7	26.6	28.5	30.4	30.5	29.5	27.2	24.1	20.8	(29)
(30)			MCDWB	24.1	25.6	27.2	30.3	33.2	34.6	35.9	35.7	34.5	32.8	29.4	25.5	(30)
(31)		5%	WB	18.2	19.0	20.5	22.0	25.7	27.8	29.7	29.8	28.8	26.6	23.2	20.1	(31)
(32)			MCDWB	23.6	24.6	26.8	29.8	32.9	34.1	35.3	35.3	33.9	32.3	29.0	25.0	(32)
(33)		10%	WB	17.5	18.1	19.6	21.2	24.7	27.1	29.0	29.1	28.2	25.9	22.2	19.4	(33)
(34)			MCDWB	22.8	24.4	26.5	29.8	33.2	34.0	35.1	35.0	33.4	31.5	28.1	24.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.6	12.1	12.9	14.1	15.6	15.4	13.0	13.1	13.4	13.7	11.9	11.5	(35)
(36)			MCDBR	14.0	15.1	16.2	17.0	17.9	17.7	14.2	14.7	14.6	14.7	13.5	13.2	(36)
(37)			MCWBR	6.2	6.2	5.9	5.7	6.1	6.5	6.0	6.4	7.1	6.5	5.6	5.9	(37)
(38)		5% WB	MCDWB	11.8	12.0	12.0	14.1	15.7	14.9	12.3	12.6	13.4	13.4	12.0	11.6	(38)
(39)			MCWBR	6.6	6.0	5.9	5.7	6.1	6.4	5.8	5.9	6.6	6.4	5.8	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.450	0.503	0.570	0.609	0.596	0.637	0.731	0.705	0.600	0.505	0.459	0.441	(40)	
(41)		taud	1.999	1.856	1.704	1.653	1.668	1.623	1.513	1.558	1.710	1.891	2.012	2.031	(41)	
(42)		Ebn at Noon	790	776	745	728	734	699	637	651	712	762	774	780	(42)	
(43)		Edh at Noon	158	195	239	256	252	261	291	278	234	186	155	148	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.26	5.02	5.88	6.69	7.39	7.51	7.01	6.81	6.37	5.60	4.57	4.01	(44)	
(45)		RadStd	0.22	0.36	0.38	0.39	0.20	0.16	0.23	0.15	0.12	0.10	0.11	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.55	+0.73	N/A	N/A	N/A	N/A	N/A	N/A	+188	+186

Nomenclature: See separate page